Parth Mital

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github.com/parthmital | parthmital-portfolio.vercel.app

Education & Certifications

Vellore Institute of Technology (VIT) **2023 – 2027**
B.Tech in Computer Science & Engineering (CGPA: 8.75 / 10) *Vellore, India*

OCI 2025 Generative AI Professional **July 2026**
Oracle Certified Professional; ID: 103485063OCI25GAIOCP *Valid until July 2028*

Experience

Concept Marketing **June 2026 – July 2026**
Sole Software Developer & Maintainer (Contract) *Vadodara, India*

- Modernised a mission-critical ERP/CRM for a 60+ employee HVAC dealership, replacing an offline, single-desktop ASP.NET & SQL Server system with Next.js, TypeScript, & Prisma on Vercel, **available 24/7 on any device**.
- Migrated **114K+ records from 42 legacy tables to Supabase PostgreSQL with zero data loss** & no operational disruption, porting legacy stored procedures to parameterised SQL builders with allowlisted columns.
- Hardened internet-facing access with HMAC-signed sessions, deny-by-default role routing, nonce-based CSP, CSRF origin checks, & row-level security; automated **AES-256 encrypted daily backups** via GitHub Actions.
- Owned architecture, testing, deployment, & maintenance solo, turning feedback from staff using it 6–8 hours daily into features; now **managing 32K+ service calls**, gated in CI by 591 Vitest tests.

LearnLine Edustation (IntelGrader) **June 2026 – July 2026**
AI Engineer Intern *Remote*

- Won the off-campus offer via Wellfound with a YOLOv8s crossed-out handwriting detector, 2-phase fine-tuned on 100 self-annotated scans; OCR masking **cut token clutter by 7.11%** at 80–96% text similarity.
- Engineered a native Android capture app in Kotlin, Jetpack Compose, CameraX, & OpenCV, bridged into the ops web portal, with TFLite page segmentation & **quad-stability auto-capture over 7 stable frames**.
- Trained a multi-task MobileNetV3 scoring blur, lighting, & geometry plus 7 defect flags; float16 TFLite quantisation **shrank it 82.5% to 2.19 MB** with 2.55 ms inference latency.
- Designed concurrent question bank pipelines: PyMuPDF ingestion **extracting 1,033 questions & 383 visuals**, dual-GPU EasyOCR over 4,574 pages, & AWS Bedrock structuring with zero validation errors.
- Drove architecture & product direction with co-founders for an AI lesson page generator compiling Bedrock lesson specs into static HTML across **6 reusable archetypes**, gated by html-validate, Playwright, & axe-core checks.
- Developed ResNet50 contrastive writer embeddings over 3,946 answer pages, searching **690 clustering configurations**; prototyped Groq, Kokoro TTS, & Stable Diffusion doodle lecture videos.

Projects & Research

NeuroScan AI: Brain Tumour Benchmark & Clinical Platform ***PyTorch, FastAPI, React, VTK.js***

- Benchmarked **24 architectures (15 unique) across 3 ML pipelines** with one teammate under a faculty advisor, splitting training, platform, & writing; co-authored 2 Elsevier-format papers under faculty review.
- Trained 9 CNN & transformer backbones per 2D task over 5 folds, deduplicating 9,081 images via perceptual hashing for zero test leakage; fold ensembles reached **99.42% detection accuracy**.
- Compared 6 3D segmentation networks on 369 BraTS 2020 cases with Holm-corrected tests (UNet best, 0.8199 Dice); deployed VNet at **0.7965 Dice with 1.4M parameters & 37.5 ms latency**.
- Built a full-stack clinical platform chaining all 3 models in a background queue (**about 4 s per scan on CPU**), with VTK.js multi-planar 3D viewing, PDF reports, TOTP 2FA, & 100 tests in CI.

Technical Skills

Languages: Python, JavaScript, SQL, Java, Bash, PowerShell
AI, ML & Data Science: Scikit-learn, Hugging Face Transformers, Sentence-Transformers, timm, ONNX Runtime, LangGraph, FAISS, vLLM, XGBoost, CatBoost, RapidFuzz, Pandas, NumPy
Frameworks & Web Systems: Node.js, Tailwind CSS, shadcn/ui, TanStack Query, SQLModel, SQLite, RESTful APIs
DevOps, Cloud & Tooling: Git, Linux, CI/CD, Dependabot, Semgrep, gitleaks, pytest, Vite, Label Studio
Coursework: Data Structures & Algorithms, OS, DBMS, Computer Networks, Software Engineering, AI, NLP, Cloud Architecture, Cryptography & Network Security, Digital Image Processing, Probability & Statistics